



The University of Faisalabad

Department of Computational Sciences

Compiler Construction– 3(2-1)

Subject Code: CS-411

LAB MANUAL

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Lab No. 01: String operations: Uppercase

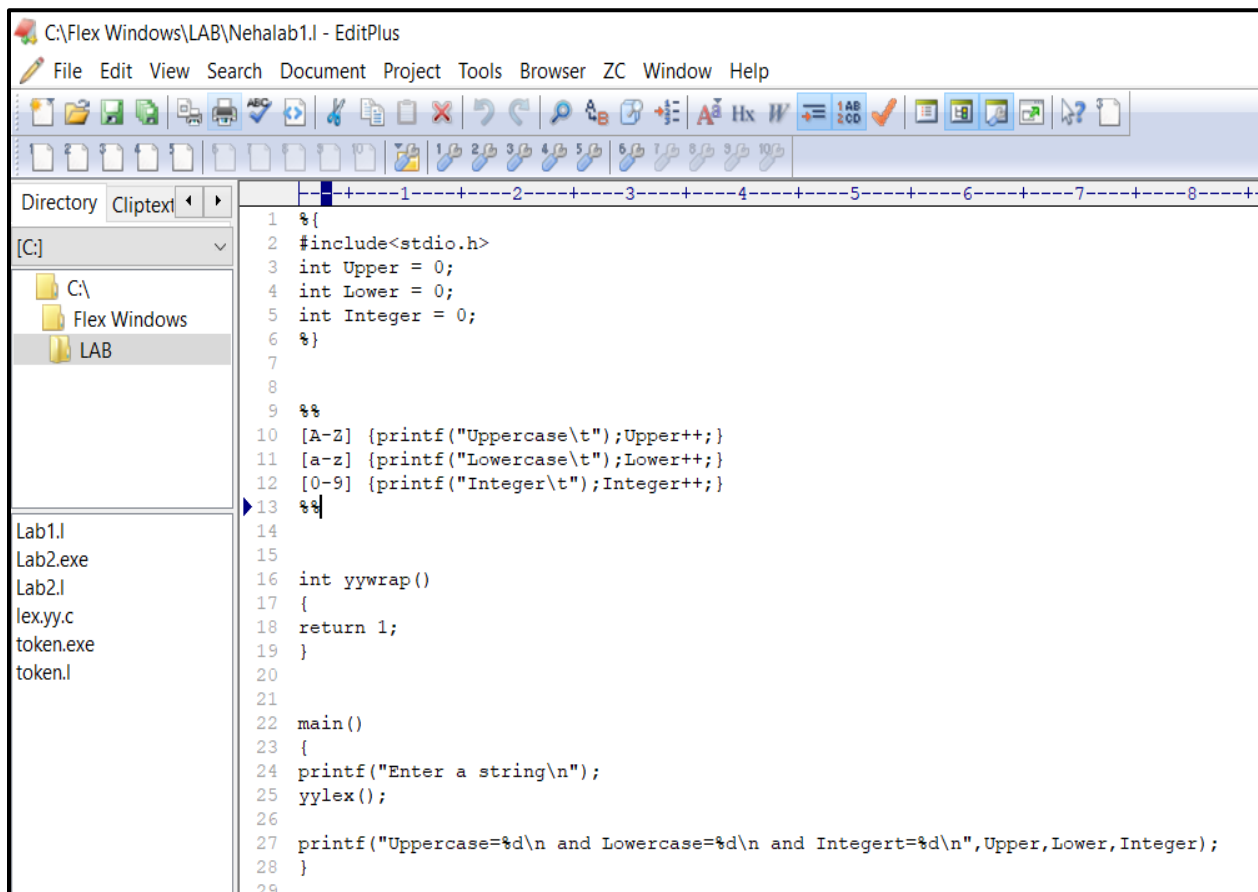
Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the concept of string operations uppercase.

Lab Work:

Program:



```
C:\Flex Windows\LAB\Nehalab1.l - EditPlus
File Edit View Search Document Project Tools Browser ZC Window Help
[Icons]
Directory Cliptext
[C:]
  C\
  Flex Windows
  LAB
Lab1.l
Lab2.exe
Lab2.l
lex.yy.c
token.exe
token.l

1  %{
2  #include<stdio.h>
3  int Upper = 0;
4  int Lower = 0;
5  int Integer = 0;
6  %}
7
8
9  %%
10 [A-Z] {printf("Uppercase\t");Upper++;}
11 [a-z] {printf("Lowercase\t");Lower++;}
12 [0-9] {printf("Integer\t");Integer++;}
13 %%
14
15
16 int yywrap()
17 {
18     return 1;
19 }
20
21
22 main()
23 {
24     printf("Enter a string\n");
25     yylex();
26
27     printf("Uppercase=%d\n and Lowercase=%d\n and Integer=%d\n",Upper,Lower,Integer);
28 }
29
```

Output:



```
C:\Flex Windows\LAB\Nehalab1.exe
Enter a string
Neha 123
Uppercase      Lowercase      Lowercase      Lowercase      Integer      Integer Integer
```

The screenshot shows a Windows command prompt window with the title bar "C:\Flex Windows\LAB\Nehalab1.exe". The prompt "Enter a string" is displayed, followed by the user input "Neha 123". The program's output is a single line: "Uppercase Lowercase Lowercase Lowercase Integer Integer Integer". The output consists of seven words, each followed by a large number of spaces, suggesting a tab-separated format. The words are "Uppercase", "Lowercase", "Lowercase", "Lowercase", "Integer", "Integer", and "Integer".

Lab No. 02: String operations: Lowercase

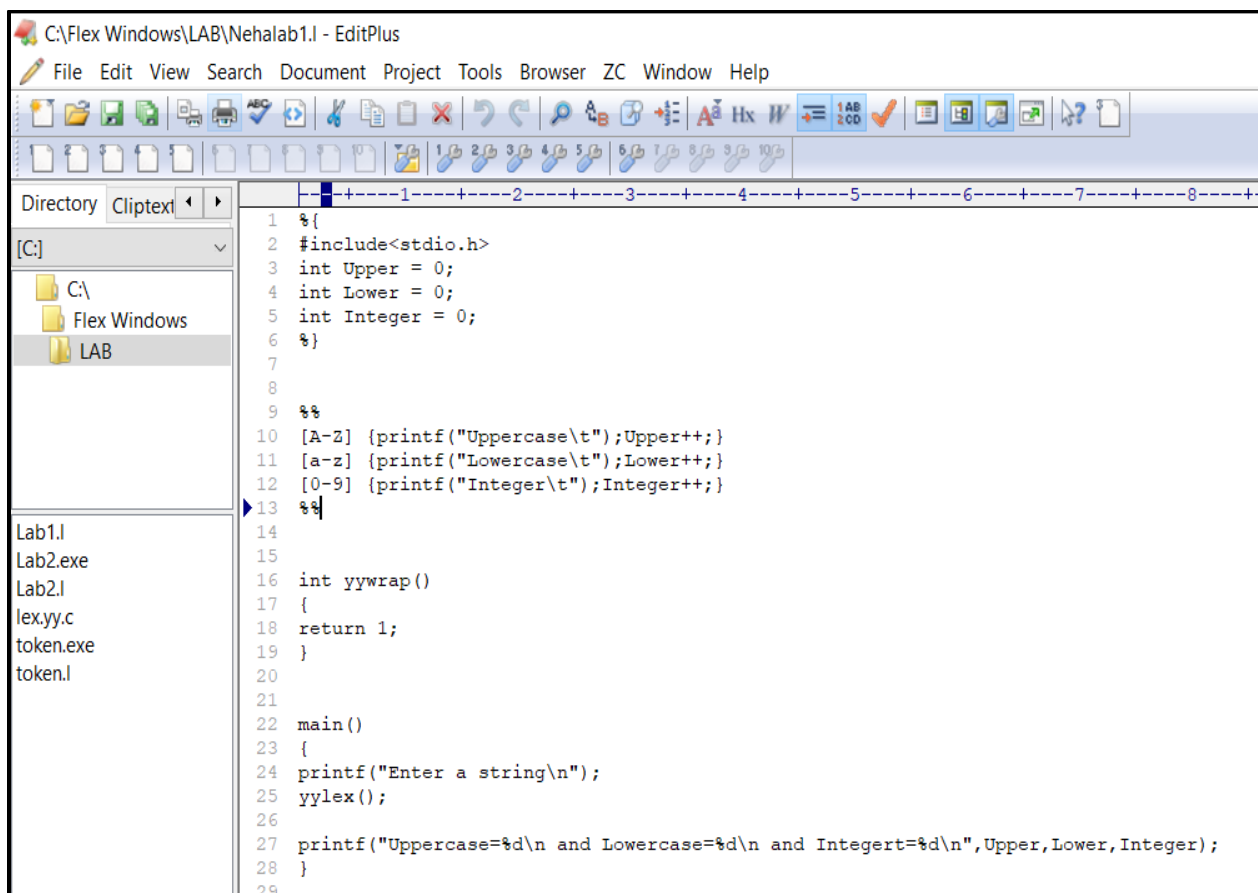
Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the concept of string operations lowercase.

Lab Work:

Program:



```
1  %{\n2  #include<stdio.h>\n3  int Upper = 0;\n4  int Lower = 0;\n5  int Integer = 0;\n6  %}\n7\n8\n9  %%\n10 [A-Z] {printf("Uppercase\\t");Upper++;}\n11 [a-z] {printf("Lowercase\\t");Lower++;}\n12 [0-9] {printf("Integer\\t");Integer++;}\n13 %%\n14\n15\n16 int yywrap()\n17 {\n18     return 1;\n19 }\n20\n21\n22 main()\n23 {\n24     printf("Enter a string\\n");\n25     yylex();\n26\n27     printf("Uppercase=%d\\n and Lowercase=%d\\n and Integer=%d\\n",Upper,Lower,Integer);\n28 }\n29
```

Output:



```
C:\Flex Windows\LAB\Nehalab1.exe
Enter a string
Neha 123
Uppercase      Lowercase      Lowercase      Lowercase      Integer      Integer Integer
```

The screenshot shows a Windows command prompt window with the title bar "C:\Flex Windows\LAB\Nehalab1.exe". The prompt "Enter a string" is displayed, followed by the user input "Neha 123". The program's output is a single line: "Uppercase Lowercase Lowercase Lowercase Integer Integer Integer". The output consists of seven words separated by spaces, with varying amounts of padding between them.

Output:



```
C:\Flex Windows\LAB\Nehalab1.exe
Enter a string
Neha 123
Uppercase      Lowercase      Lowercase      Lowercase      Integer      Integer Integer
```

The screenshot shows a Windows command prompt window with the title bar "C:\Flex Windows\LAB\Nehalab1.exe". The prompt "Enter a string" is displayed, followed by the user input "Neha 123". The program then outputs the string "Neha 123" with each character analyzed: 'N' is Uppercase, 'e', 'h', 'a' are Lowercase, ' ' is a space, '1', '2', '3' are Integers. The output is formatted as "Uppercase Lowercase Lowercase Lowercase Integer Integer Integer".

Lab No. 04: String operations: Vowel

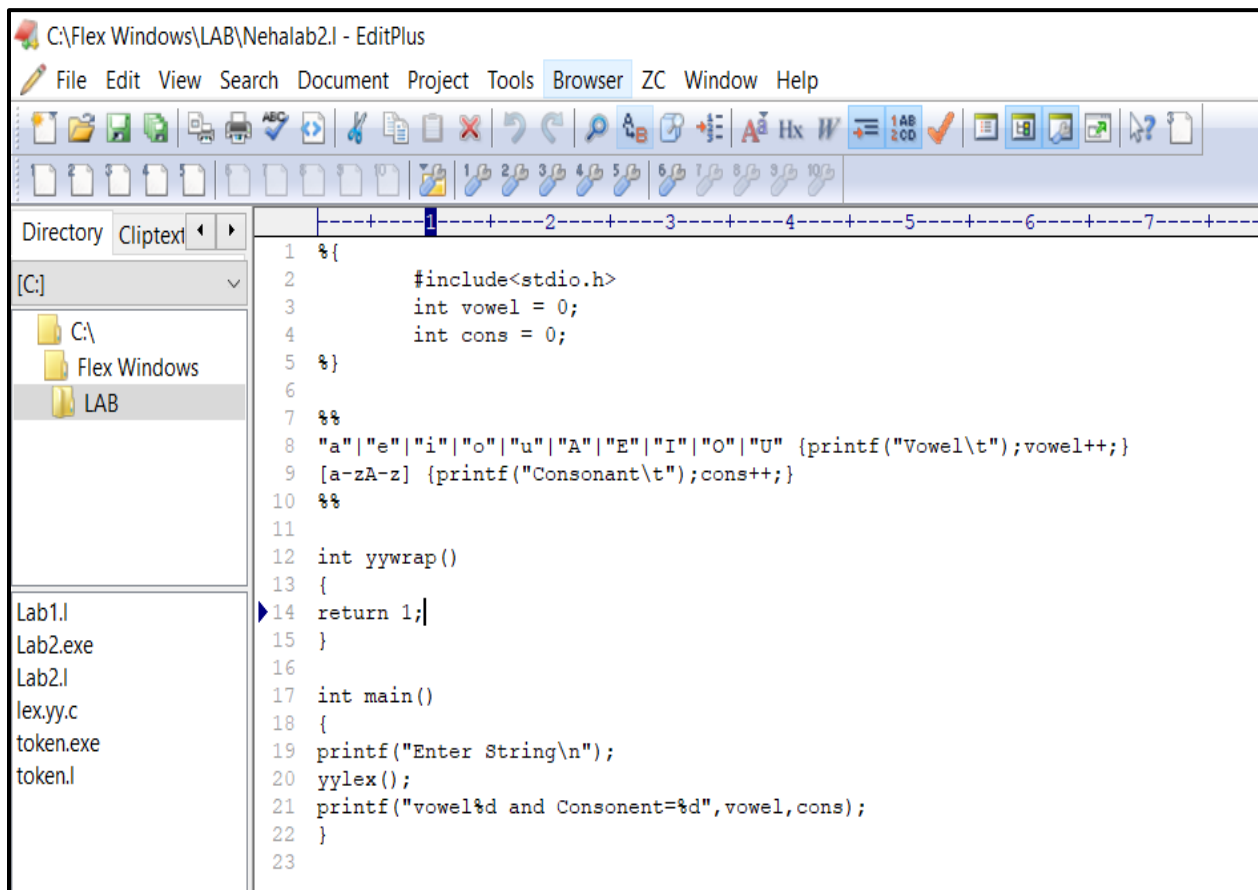
Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the concept of string operations vowel.

Lab Work:

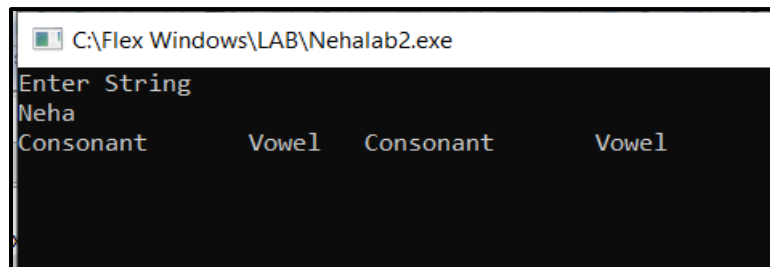
Program:



The screenshot shows the EditPlus IDE interface. The title bar indicates the file path is C:\Flex Windows\LAB\Nehalab2.l. The menu bar includes File, Edit, View, Search, Document, Project, Tools, Browser, ZC, Window, and Help. The toolbar contains various icons for file operations, editing, and formatting. The left sidebar shows the directory structure: [C:] > Flex Windows > LAB. The file list includes Lab1.l, Lab2.exe, Lab2.l, lex.yy.c, token.exe, and token.l. The main editor window displays the following C code:

```
1  %{
2      #include<stdio.h>
3      int vowel = 0;
4      int cons = 0;
5  %}
6
7  %%
8  "a"|"e"|"i"|"o"|"u"|"A"|"E"|"I"|"O"|"U" {printf("Vowel\t");vowel++;}
9  [a-zA-z] {printf("Consonant\t");cons++;}
10 %%
11
12 int yywrap()
13 {
14     return 1;
15 }
16
17 int main()
18 {
19     printf("Enter String\n");
20     yylex();
21     printf("vowel%d and Consonent=%d",vowel,cons);
22 }
23
```

Output:



```
C:\Flex Windows\LAB\Nehalab2.exe
Enter String
Neha
Consonant      Vowel  Consonant      Vowel
```

The screenshot shows a Windows command prompt window with the title bar "C:\Flex Windows\LAB\Nehalab2.exe". The prompt "Enter String" is displayed, followed by the user input "Neha". The program's output is "Consonant Vowel Consonant Vowel", where the words are separated by spaces and the first two words are aligned under the input string.

Lab No. 05: String operations: Consonant

Learning Objectives / Outcomes:

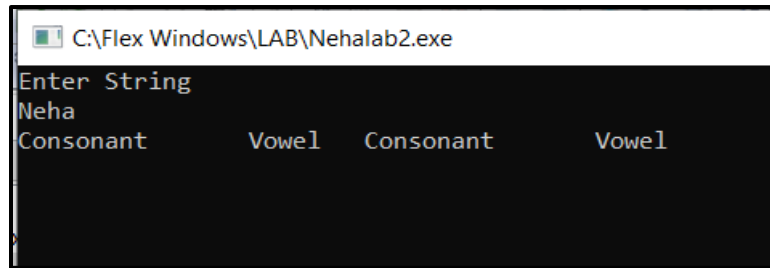
After completion of this Lab, students will be able:

- To understand the concept of string operations consonant.

Lab Work:

Program:

Output:



```
C:\Flex Windows\LAB\Nehalab2.exe
Enter String
Neha
Consonant      Vowel  Consonant      Vowel
```

The screenshot shows a Windows command prompt window with the title bar "C:\Flex Windows\LAB\Nehalab2.exe". The prompt "Enter String" is displayed, followed by the user input "Neha". The program then outputs the classification of each character: "Consonant" for 'N', "Vowel" for 'e', "Consonant" for 'h', and "Vowel" for 'a'.

Lab No. 06: Token Generations

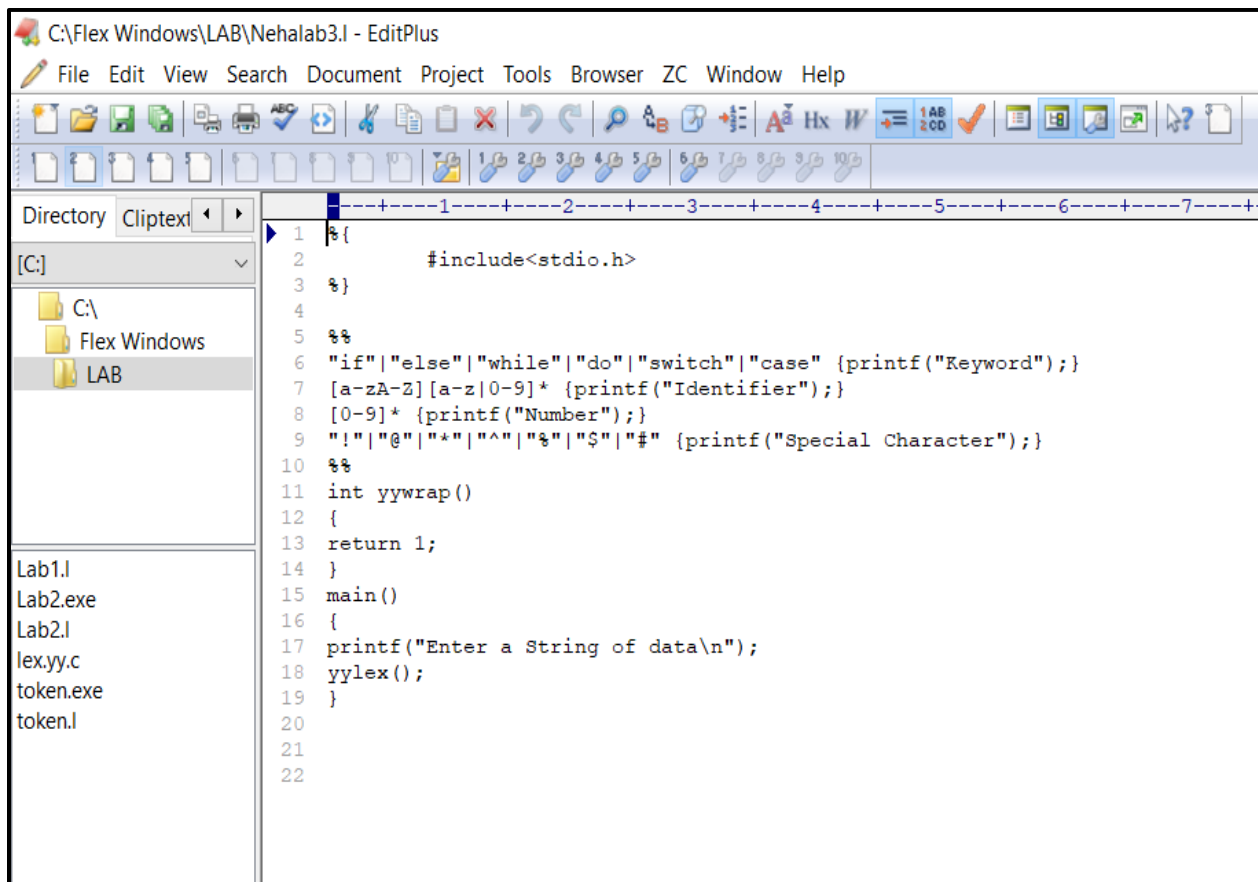
Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the concept of Token Generations.

Lab Work:

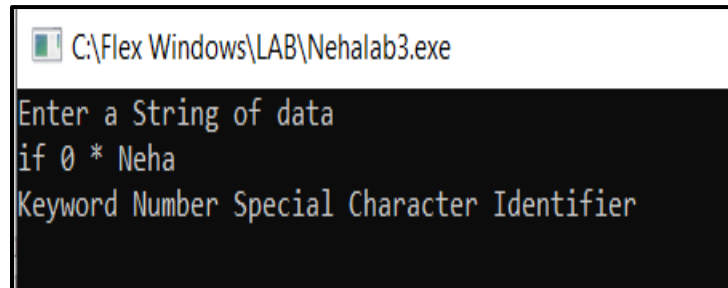
Program:



The screenshot shows the EditPlus IDE interface. The title bar reads "C:\Flex Windows\LAB\Nehalab3.I - EditPlus". The menu bar includes File, Edit, View, Search, Document, Project, Tools, Browser, ZC, Window, and Help. The toolbar contains various icons for file operations, editing, and development. The left sidebar shows the "Directory" pane with the file explorer set to "C:\", displaying folders "Flex Windows" and "LAB", and a list of files: "Lab1.l", "Lab2.exe", "Lab2.l", "lex.yy.c", "token.exe", and "token.l". The main editor window displays a C program with the following code:

```
1  %{
2      #include<stdio.h>
3  %}
4
5  %%
6  "if"|"else"|"while"|"do"|"switch"|"case" {printf("Keyword");}
7  [a-zA-Z][a-z|0-9]* {printf("Identifier");}
8  [0-9]* {printf("Number");}
9  "!"|"@"|"*"|"^"|"%"|"$"|"#" {printf("Special Character");}
10 %%
11 int yywrap()
12 {
13     return 1;
14 }
15 main()
16 {
17     printf("Enter a String of data\n");
18     yylex();
19 }
20
21
22
```

Output:



```
C:\Flex Windows\LAB\Nehalab3.exe
Enter a String of data
if 0 * Neha
Keyword Number Special Character Identifier
```

Lab No. 07: Symbol Table

Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the concept of Symbol Table .

Lab Work:

ALGORITHM:

1. Start the Program.
2. Get the input from the user with the terminating symbol '\$'.
3. Allocate memory for the variable by dynamic memory allocation function.
4. If the next character of the symbol is an operator then only the memory is allocated.
5. While reading, the input symbol is inserted into symbol table along with its memory address.
6. The steps are repeated till '\$' is reached.
7. To reach a variable, enter the variable to the searched and symbol table has been checked for corresponding variable, the variable along its address is displayed as result.
8. Stop the program.

Code:

```
#include<stdio.h>

#include<conio.h>

#include<malloc.h>

#include<string.h>

#include<math.h>

#include<ctype.h>

void main()

{

int i=0,j=0,x=0,n,flag=0; void *p,*add[15];

char ch,srch,b[15],d[15],c;

//clrscr();

printf("expression terminated by $:");

while((c=getchar())!='$')

{

b[i]=c; i++;

}

n=i-1;

printf("given expression:");

i=0;

while(i<=n)

{
```

```

printf("%c",b[i]); i++;
}
printf("symbol table\n");
printf("symbol\taddr\ttype\n");
while(j<=n)
{
c=b[j]; if(isalpha(toascii(c)))
{
if(j==n)
{
p=malloc(c); add[x]=p;
d[x]=c;
printf("%c\t%d\tidentifier\n",c,p);
}
else
{
ch=b[j+1];
if(ch=='+' || ch=='-' || ch=='*' || ch=='=')
{
p=malloc(c);
add[x]=p;
d[x]=c;

```

```

printf("%c\t%d\tidentifier\n",c,p);

x++;

}

}

}j++;

}

printf("the symbol is to be searched\n");

srch=getch();

for(i=0;i<=x;i++)

{

if(srch==d[i])

{

printf("symbol found\n");

printf("%c%s%d\n",srch,"@address",add[i]);

flag=1;

}

}

if(flag==0)

printf("symbol not found\n");

//getch();

```

Output:

```
expression terminated by $:a+b*c=d$
given expression:a+b*c=d$
symbol  addr  type
a       1892  identifier
b       1994  identifier
c       2096  identifier
d       2200  identifier
the symbol is to be searched
-
```

Lab No. 08: Three Address code

Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the concept of Three Address code.

Lab Work:

Problem 01:

Generate three address code for the following code: $c = 0$

```
do
{
if (a < b) then
x++;
else
x--;
c++;
} while (c < 5)
```

Solution-

Three address code for the given code is-

1. $c = 0$
2. if (a < b) goto (4)
3. goto (7)
4. $T1 = x + 1$
5. $x = T1$
6. goto (9)
7. $T2 = x - 1$
8. $x = T2$
9. $T3 = c + 1$
10. $c = T3$
11. if (c < 5) goto (2)

Problem-02:

Generate three address code for the following codewhile

```
(A < C and B > D) do
if A = 1 then C = C + 1
else
while A <= D
do A = A + B
```

Solution-

Three address code for the given code is-

1. if (A < C) goto (3)
2. goto (15)
3. if (B > D) goto (5)
4. goto (15)
5. if (A = 1) goto (7)
6. goto (10)
7. T1 = c + 1
8. c = T1
9. goto (1)
10. if (A <= D) goto (12)
11. goto (1)
12. T2 = A + B
13. A = T2
14. goto (10)

Problem-03:

Generate three address code for the following codeswitch

```
(ch)
{
case 1 : c = a + b;
break;
case 2 : c = a - b;
break;
}
```

Solution-

Three address code for the given code isif

```
ch = 1 goto L1
if ch = 2 goto L2
L1:
T1 = a + b
c = T1
goto Last
L2:
T1 = a - b
c = T2
goto Last
Last:
```

Lab No. 09: Directed Acyclic Graph

Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the concept of Directed Acyclic Graph.

Lab Work:

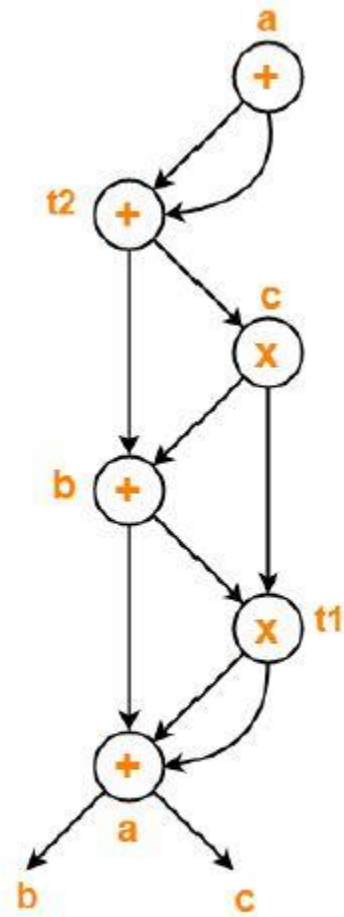
Problem:

Construct a DAG for the following three address code-

1. $a = b + c$
2. $t1 = a \times a$
3. $b = t1 + a$
4. $c = t1 \times b$
5. $t2 = c + b$
6. $a = t2 + t2$

Solution:

Directed acyclic graph for the given three address code is-



Directed Acyclic Graph

Lab No. 10: Three Address code II

Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the concept of Three Address code II.

Lab Work:

Problem:

Consider the following codeprod

```
= 0 ;  
i = 1 ;  
do  
{  
  prod = prod + a[ i ] x b[ i ] ;  
  i = i + 1 ;  
} while (i <= 10) ;  
1. Compute the three address code.  
2. Compute the basic blocks and draw the flow graph.
```

Solution:

Part-01

Three address code for the given code isprod

```
= 0  
i = 1  
T1 = 4 x i  
T2 = a[T1]  
T3 = 4 x i  
T4 = b[T3]  
T5 = T2 x T4  
T6 = T5 + prod  
prod = T6  
T7 = i + 1  
i = T7  
if (i <= 10) goto (3)
```

Lab No. 11: Basic Blocks

Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the concept of Basic Blocks.

Lab Work:

Problem:

Consider the following codeprod

```
= 0 ;  
i = 1 ;  
do  
{  
  prod = prod + a[ i ] x b[ i ] ;  
  i = i + 1 ;  
} while (i <= 10) ;
```

1. Compute the three address code.
2. Compute the basic blocks and draw the flow graph.

Part-02:

Step-01:

We identify the leader statements asprod

= 0 is a leader because first statement is a leader.

T1 = 4 x i is a leader because target of conditional or unconditional goto is a leader.

Step-02:

The above generated three address code can be partitioned into 2 basic blocks as

```
prod = 0  
i = 1
```

Block-1

```
T1 = 4 x i  
T2 = a[T1]  
T3 = 4 x i  
T4 = b[T3]  
T5 = T2 x T4  
T6 = T5 + prod  
prod = T6  
T7 = i + 1  
i = T7  
if (i <= 10) goto (B2)
```

Block-2

Basic Blocks

Lab No. 12: Flow Graph

Learning Objectives / Outcomes:

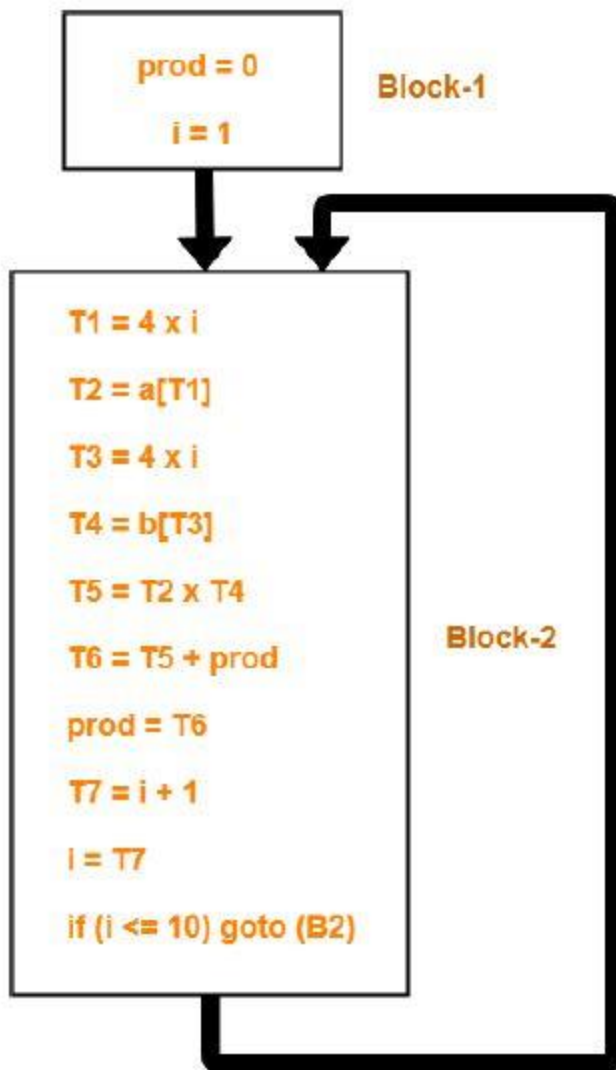
After completion of this Lab, students will be able:

- To understand the concept of Flow Graph.

Lab Work:

Step-03:

The flow graph is-



Flow Graph

To gain better understanding about these Miscellaneous Problems,

Lab No. 13: Data Structure for Symbol Table

Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the concept of Data Structure for Symbol Table.

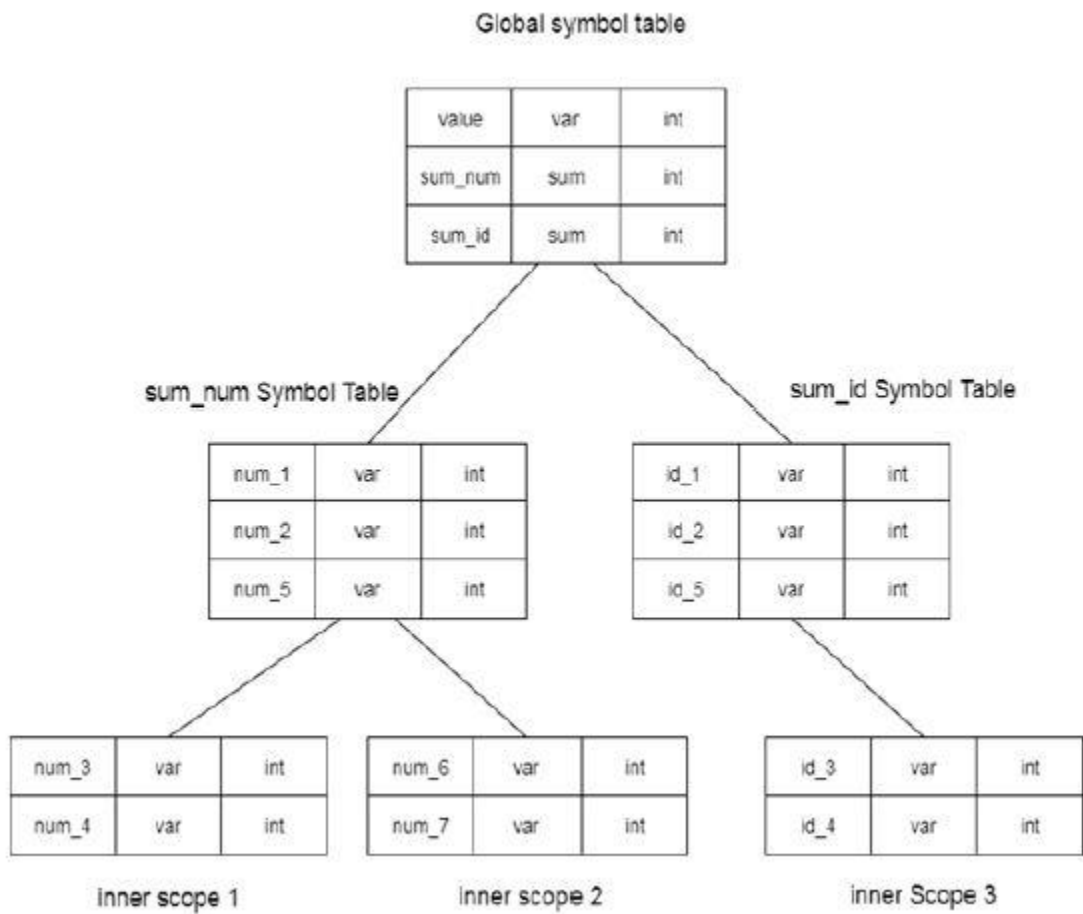
Lab Work:

Global symbol table can be accessed by all the procedures and scope symbol table.

The scope of a name and symbol table is arranged in the hierarchy structure as shown below:

```
int value=10;
void sum_num()
{
    int num_1;
    int num_2;
    {
        int num_3;
        int num_4;
    }
    int num_5;
    {
        int_num 6;
        int_num 7;
    }
}
Void sum_id
{
    int id_1;
    int id_2;
    {
        int id_3;
        int id_4;
    }
    int num_5;}
```

The above grammar can be represented in a hierarchical data structure of symbol tables:



The global symbol table contains one global variable and two procedure names. The name mentioned in the sum_num table is not available for sum_id and its child tables

Lab No. 14: Practice

Learning Objectives / Outcomes:

Lab Work:

Lab No. 15: Practice

Learning Objectives / Outcomes:

Lab Work:

Lab No. 16: Quiz

Learning Objectives / Outcomes:

Lab Work: